

Full-Stack SaaS with Google-Powered Free Tools

A complete student roadmap — Research → Design → Code → Deploy → AI

Gemini · Jules · Stitch · NotebookLM · Firebase · Vertex AI · Gemma · Cloud Run

Yeh guide un students ke liye hai jo ek complete SaaS product banana chahte hain — bina koi paisa kharch kiye — sirf Google ke free tools use karke. Har phase mein kya karna hai, kaunsa tool use karna hai, aur actual commands/code sab included hai.

Google Free Tools — Quick Reference

Tool	Kaam	Free Tier
Gemini 2.5 Flash	Main LLM — code, chat, analysis	Google AI Studio — free
Gemma 3 (local)	Open-source LLM locally chalao	Ollama ya Kaggle free
Jules (Google)	AI coding agent — GitHub issues solve karta hai	Beta — free
Stitch (Google)	UI/UX design → React code generate karta hai	Beta — free
NotebookLM	Research, PDF summarize, podcast banaao	Free
Firebase	Auth + Firestore DB + Hosting	Generous free tier
Google Cloud Run	Backend deploy karo serverless	Free tier available
Vertex AI	Production AI APIs	Free credits on signup
Google Colab	Python/ML experiments	Free GPU
Gemini CLI	Terminal se AI commands chalao	Free with API key
IDX (Project IDX)	Browser-based full-stack IDE	Free

Phase 0 — Setup & Mindset

Pehle Yeh Samjho

SaaS (Software as a Service) matlab ek web application jo users subscription ya free model pe use karte hain. Tumhara goal: ek real problem solve karna, working product banana, aur deploy karna — sab kuch free Google tools se.

Ek SaaS Product ke 5 Core Parts

- **Frontend:** Jo user dekhta hai — React/Next.js se banaoge
- **Backend:** Business logic + APIs — Node.js/Python
- **Database:** Data store karne ke liye — Firestore ya Supabase
- **Auth:** Login/signup system — Firebase Auth
- **AI Layer:** Smart features — Gemini API se

Environment Setup — Ek Baar Karo

1. Google Account + API Key lao

aistudio.google.com pe jao → API Key generate karo. Yeh tumhara main key hai.

```
GEMINI_API_KEY=your_key_here
```

2. Node.js + npm install karo

nodejs.org se download karo. LTS version lo.

```
node --version # v20+ chahiye npm --version
```

3. Gemini CLI install karo

Terminal se Google ke saath directly baat karo:

```
npm install -g @google/gemini-cli gemini # terminal mein AI agent start hoga
```

4. Firebase CLI install karo

```
npm install -g firebase-tools firebase login firebase init
```

5. Project IDX use karo (optional but recommended)

idx.dev pe jao — yeh Google ka browser-based VS Code jaisa IDE hai jo Firebase, Cloud Run sab kuch pre-configured deta hai. Local setup ki zaroorat nahi.

Phase 1 — Research & Idea Validation

Tool: NotebookLM

notebooklm.google.com — Yahan tum PDFs, articles, URLs upload karo aur AI se questions poochho. Research ke liye best tool hai.

Kaise Use Karo

1. NotebookLM open karo, new notebook banao
2. Apni domain ki 5-10 relevant PDFs ya articles add karo (research papers, competitor analysis)
3. Prompt do: 'Is problem space mein kya gaps hain? Kaunse users underserved hain?'
4. Audio Overview feature use karo — AI ek podcast-style summary banata hai
5. Key insights note karo aur apna SaaS idea finalize karo

Gemini se Idea Validate Karo

aistudio.google.com pe jao, yeh prompt use karo:

Mera SaaS idea hai: [TUMHARA IDEA]. Mujhe batao: 1. Is problem ka market size kya hai? 2. 3 competitors kaun hain aur unki limitations kya hain? 3. Mera unique value proposition kya ho sakta hai? 4. Technical feasibility assessment karo 5. MVP features ki list do

Deliverable after Phase 1

Ek 1-page document: Problem statement, target user, 5 core features, aur yeh kyun banayenge. Aage badho tabhi jab yeh clear ho.

Phase 2 — UI/UX Design

Tool: Stitch (labs.google/stitch)

Stitch Google ka naya experimental tool hai jo natural language se UI banata hai aur React/HTML code generate karta hai. Designer hona zaroori nahi.

Stitch Workflow

Step 1 — Describe karo

Prompt do: 'A SaaS dashboard for [TUMHARA PRODUCT] with sidebar navigation, stats cards, and a data table. Clean minimal design.'

Step 2 — Iterate karo

Jo design aaye usse modify karo: 'Make the sidebar darker, add a search bar at top, use Google's Material Design style'

Step 3 — Export karo

React ya HTML code copy karo. Yeh directly tumhare project mein paste ho sakta hai

Step 4 — Screens banao

Dashboard, Login page, Settings, Landing page — har screen ke liye alag prompt do

Stitch Prompts jo Kaam Karte Hain

- "Landing page for a [PRODUCT] SaaS with hero section, features grid, pricing table, and CTA buttons"
- "User authentication forms: login, signup, forgot password — minimal and clean"
- "Settings page with profile section, notification toggles, and danger zone"
- "Pricing page with 3 tiers: Free, Pro \$9/mo, Enterprise — highlight the middle one"

Agar Stitch Access Na Ho

Gemini ko directly prompt karo: 'Write complete React + Tailwind CSS code for a [SCREEN NAME] page for [PRODUCT]. Include all components, responsive design, and dark mode support.' Ya v0.dev (Vercel ka free tool) use karo for UI generation.

Phase 3 — Frontend Development

Stack: Next.js 14 + Tailwind CSS + shadcn/ui

Next.js Google-friendly hai, SEO ke liye best hai, aur full-stack bhi ban sakta hai. shadcn/ui ready-made components deta hai. Sab free.

Project Create Karo

```
npx create-next-app@latest my-saas # Options: TypeScript=Yes, Tailwind=Yes, App Router=Yes, src/=Yes cd my-saas npx shadcn@latest init npm run dev
```

Jules se Code Generate Karo (jules.google.com)

Jules Google ka AI coding agent hai jo GitHub integration ke saath kaam karta hai. GitHub repo connect karo, issue create karo, Jules automatically code likhta aur PR banata hai.

Jules Use Karna

1. jules.google.com pe jao, Google account se login karo
2. GitHub repo connect karo (tumhara Next.js project)
3. GitHub issue create karo jaise: 'Create a responsive dashboard layout with sidebar, header with user avatar, and main content area using Tailwind CSS'
4. Jules analyze karega aur ek pull request banayega with actual code
5. Review karo, merge karo — done!

Ya Gemini CLI se Code Likhao

Project folder mein terminal kholo:

```
cd my-saas gemini # Ab Gemini ko instructions do: > Create a complete dashboard page at src/app/dashboard/page.tsx > It should have: stats cards (users, revenue, growth), > a recent activity table, and use shadcn Card components > Make it fully responsive with Tailwind CSS
```

Key Frontend Files Structure

```
src/ app/ (auth)/login/page.tsx # Login page (auth)/signup/page.tsx # Signup page dashboard/page.tsx # Main dashboard dashboard/settings/page.tsx # Settings api/ # API routes (backend) components/ ui/ # shadcn components layout/Sidebar.tsx # Sidebar layout/Header.tsx # Header lib/ firebase.ts # Firebase config gemini.ts # AI helper functions
```

Gemini API — Frontend Integration

```
npm install @google/generative-ai
```

src/lib/gemini.ts file banao:

```
import { GoogleGenerativeAI } from '@google/generative-ai'; const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY!); export async function askGemini(prompt: string): Promise { const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash' }); const result = await model.generateContent(prompt); return result.response.text(); }
```

Phase 4 — Backend & APIs

Option A — Next.js API Routes (Simplest)

Next.js mein backend aur frontend ek hi project mein hota hai. `src/app/api/` folder mein API routes banao. Isse alag server ki zaroorat nahi — deploy bhi saath mein hoga.

Example: AI Chat API Route

`src/app/api/chat/route.ts`:

```
import { GoogleGenerativeAI } from '@google/generative-ai'; import { NextRequest, NextResponse } from 'next/server'; const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY!); export async function POST(req: NextRequest) { const { message, history } = await req.json(); const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash' }); const chat = model.startChat({ history }); const result = await chat.sendMessage(message); return NextResponse.json({ reply: result.response.text() }); }
```

Option B — Python FastAPI Backend (Advanced)

Agar ML processing, complex logic, ya Python libraries chahiye toh alag FastAPI backend banao aur Google Cloud Run pe deploy karo.

FastAPI Setup

```
pip install fastapi uvicorn google-generativeai python-dotenv # main.py from fastapi import FastAPI from fastapi.middleware.cors import CORSMiddleware import google.generativeai as genai import os app = FastAPI() app.add_middleware(CORSMiddleware, allow_origins=['*']) genai.configure(api_key=os.environ['GEMINI_API_KEY']) @app.post('/api/analyze') async def analyze(data: dict): model = genai.GenerativeModel('gemini-2.0-flash') result = model.generate_content(data['prompt']) return {'result': result.text} # Run: uvicorn main:app --reload
```

Gemini Advanced Features for Backend

- **Structured Output**: JSON format mein response lo — database mein directly store karo
- **Function Calling**: Gemini ko tools do — weather check, database query etc
- **File API**: PDFs aur images process karo — Gemini ko directly bhejo
- **Context Caching**: Badi documents cache karo — cost bachao aur speed badhao
- **Grounding**: Google Search se real-time data add karo responses mein

Structured Output Example

```
# Gemini se guaranteed JSON lo import google.generativeai as genai import json model = genai.GenerativeModel('gemini-2.0-flash', generation_config={'response_mime_type': 'application/json'}) response = model.generate_content('Extract: name, email, company from this text: ' + user_text) data = json.loads(response.text) # Always valid JSON
```

Phase 5 — Database & Authentication

Firestore — Google Ka Free Database

Firestore free tier mein milta hai: Firestore (NoSQL DB), Authentication (login system), Storage (files), Hosting — sab ek saath. Students ke liye perfect.

Firestore Setup

```
# 1. console.firebase.google.com pe project banao # 2. Web app add karo, config copy karo npm
install firebase # src/lib/firebase.ts import { initializeApp } from 'firebase/app'; import {
getAuth } from 'firebase/auth'; import { getFirestore } from 'firebase/firestore'; const
firebaseConfig = { apiKey: process.env.NEXT_PUBLIC_FIREBASE_API_KEY, authDomain:
'your-app.firebaseio.com', projectId: 'your-project-id', }; const app =
initializeApp(firebaseConfig); export const auth = getAuth(app); export const db =
getFirestore(app);
```

Authentication — Google Login Add Karo

```
import { signInWithPopup, GoogleAuthProvider } from 'firebase/auth'; import { auth } from
'@lib/firebase'; export async function loginWithGoogle() { const provider = new
GoogleAuthProvider(); const result = await signInWithPopup(auth, provider); return result.user;
// user ka data mil gaya }
```

Firestore — Data Save aur Read Karo

```
import { collection, addDoc, getDocs, query, where } from 'firebase/firestore'; import { db }
from '@lib/firebase'; // Data save karo async function saveUserData(userId: string, data:
object) { await addDoc(collection(db, 'users'), { userId, ...data, createdAt: new Date() }); }
// Data read karo async function getUserData(userId: string) { const q = query(collection(db,
'users'), where('userId', '==', userId)); const snapshot = await getDocs(q); return
snapshot.docs.map(doc => doc.data()); }
```

Firestore Free Limits (Spark Plan)

- ✓ Firestore: 1 GB storage, 50K reads/day, 20K writes/day
- ✓ Auth: Unlimited users
- ✓ Storage: 5 GB
- ✓ Hosting: 10 GB storage, 360 MB/day transfer

Alternative: Supabase (PostgreSQL)

Agar SQL database chahiye toh supabase.com use karo — PostgreSQL based, free tier mein 500 MB storage, 2 projects. Firestore jaisa easy setup.

Phase 6 — AI & LLM Integration

Gemini 2.5 Flash — Primary LLM

Gemini 2.5 Flash Google AI Studio pe FREE hai. Fast hai, smart hai, 1M token context window hai. Yeh tumhara main AI engine hoga.

Common AI Features jo SaaS mein Add Kar Sakte Ho

- **AI Chat / Chatbot:** Users ke saath conversation — customer support, Q&A;
- **Document Analysis:** PDF/file upload karo, AI summarize ya analyze kare
- **Smart Search:** Natural language se search — SQL ya Firestore query auto-generate
- **Content Generation:** Emails, reports, descriptions auto-likhna
- **Data Extraction:** Unstructured text se structured data nikalna
- **Recommendation Engine:** User behavior se personalized suggestions

AI Chat Feature — Complete Implementation

```
# Frontend: src/components/AiChat.tsx # Backend: src/app/api/chat/route.ts (pehle se banaya) //
React component const [messages, setMessages] = useState([]); const [input, setInput] =
useState(''); async function sendMessage() { const userMsg = { role: 'user', parts: [{ text:
input }] }; setMessages(prev => [...prev, { role: 'user', content: input }]); const res = await
fetch('/api/chat', { method: 'POST', body: JSON.stringify({ message: input, history: messages
}), }); const { reply } = await res.json(); setMessages(prev => [...prev, { role: 'assistant',
content: reply }]); }
```

Document Analysis — PDF Upload + AI

```
// File upload + Gemini File API import { GoogleAIFileManager } from
'@google/generative-ai/server'; async function analyzeDocument(file: File, question: string) {
const fileManager = new GoogleAIFileManager(process.env.GEMINI_API_KEY!); // File upload karo
Gemini ke servers pe const buffer = await file.arrayBuffer(); const uploadResult = await
fileManager.uploadFile(buffer, { mimeType: file.type, displayName: file.name, }); // AI se
question poochho const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash' }); const
result = await model.generateContent([ { fileData: { fileUri: uploadResult.file.uri, mimeType:
file.type } }, question, ]); return result.response.text(); }
```

Gemma — Local LLM (No API Cost)

Agar API calls ki limit ho ya sensitive data ho toh Gemma locally run karo. Gemma Google ka open-source model hai — Ollama ke through free mein chalao.

```
# Ollama install karo: ollama.ai ollama pull gemma3 # 3B model download karo ollama run gemma3 #
Test karo # Python se use karo import ollama response = ollama.chat(model='gemma3', messages=[
{'role': 'user', 'content': 'Explain this code: ...'} ]) print(response['message']['content'])
```

Vertex AI — Production ke liye

Jab app real users ke saath grow kare, Vertex AI pe move karo. Google Cloud Console pe \$300 free credits milte hain signup pe. Same Gemini models, enterprise-grade reliability.

Phase 7 — Deployment

Frontend Deploy: Vercel (Recommended)

Next.js ke liye Vercel best option hai. Free tier mein unlimited personal projects. GitHub se connect karo — har push pe automatic deploy.

```
# GitHub pe code push karo git init && git add . && git commit -m 'initial' git push origin main
# Vercel pe: # 1. vercel.com pe jao, GitHub se login karo # 2. 'New Project' → apna repo select
karo # 3. Environment variables add karo (GEMINI_API_KEY, Firebase keys) # 4. Deploy! — tumhe ek
.vercel.app URL milega
```

Firebase Hosting (Alternative)

```
npm run build firebase deploy --only hosting # tumhe yourapp.web.app URL milega — free SSL
included
```

Backend Deploy: Google Cloud Run

Python FastAPI ya Node.js backend ke liye Cloud Run use karo. Serverless hai — traffic nahi toh cost nahi. Free tier: 2M requests/month.

```
# Dockerfile banao FROM python:3.11-slim WORKDIR /app COPY requirements.txt . RUN pip install -r
requirements.txt COPY . . CMD ['uvicorn', 'main:app', '--host', '0.0.0.0', '--port', '8080'] #
Deploy karo gcloud run deploy my-backend \ --source . \ --region us-central1 \
--allow-unauthenticated \ --set-env-vars GEMINI_API_KEY=your_key
```

Environment Variables — Kaise Manage Karo

KABHI BHI API keys code mein hardcode mat karo. .env.local file banao:

```
# .env.local (gitignore mein add karo!) GEMINI_API_KEY=AIZA...
NEXT_PUBLIC_FIREBASE_API_KEY=AIZA... NEXT_PUBLIC_FIREBASE_PROJECT_ID=my-project # .gitignore
mein add karo: .env.local .env
```

Phase 8 — Advanced AI Features

Gemini Function Calling — AI ko Tools do

Function calling se Gemini tumhare code ki functions ko call kar sakta hai. Matlab AI sach mein actions le sakta hai — database query, email send, calculations.

```
const tools = [{ functionDeclarations: [{ name: 'get_user_data', description: 'Get user information from database', parameters: { type: 'OBJECT', properties: { userId: { type: 'STRING', description: 'User ID' } }, required: ['userId'] } } ] }]; const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash', tools }); // Ab Gemini khud decide karega kab get_user_data call karna hai
```

RAG — Apna Knowledge Base Banao

RAG (Retrieval Augmented Generation) matlab AI ko tumhare docs ka knowledge do. Users tumhare product ke baare mein questions pooch sakte hain.

Step 1 — Documents prepare karo

Apne docs/FAQs ko chunks mein split karo (500 words each)

Step 2 — Embeddings banao

Gemini Embedding API se har chunk ka vector nikalo aur Firestore mein save karo

Step 3 — Search implement karo

User query ka embedding nikalo, similar chunks dhundo

Step 4 — Context add karo

Related chunks Gemini ke prompt mein add karo — ab accurate answers aayenge

Embedding Example

```
import { GoogleGenerativeAI } from '@google/generative-ai'; const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY!); async function getEmbedding(text: string) { const model = genAI.getGenerativeModel({ model: 'text-embedding-004' }); const result = await model.embedContent(text); return result.embedding.values; // 768-dim vector } // Cosine similarity se compare karo chunks: function cosineSim(a: number[], b: number[]) { const dot = a.reduce((sum, ai, i) => sum + ai * b[i], 0); const magA = Math.sqrt(a.reduce((sum, ai) => sum + ai*ai, 0)); const magB = Math.sqrt(b.reduce((sum, bi) => sum + bi*bi, 0)); return dot / (magA * magB); }
```

NotebookLM se RAG Content Create Karo

NotebookLM mein apne product ke saare docs add karo. Wahan se structured Q&A; pairs export karo. Yeh pairs tumhare RAG knowledge base ka foundation banenge. Smart shortcut!

Phase 9 — Complete Project Example

Example: AI Resume Analyzer SaaS

Let's build: Users resume upload karein, AI feedback de, companies listing browse karein. Yeh ek realistic SaaS idea hai — complete tech breakdown:

Component	Technology	Kaam
Landing Page	Next.js + Stitch generated	Conversion ke liye CTA, features
Auth	Firebase Google Login	1-click signup with Google
Resume Upload	Firebase Storage	PDF/DOC store karo
AI Analysis	Gemini 2.0 Flash + File API	Resume parse karo, feedback do
Dashboard	Next.js + Firestore	Past analyses, scores dikhao
Job Match	Gemini + Structured Output	Resume se job recommendations
Backend APIs	Next.js API Routes	Sab kuch same project mein
Deploy	Vercel + Firebase Hosting	Free automatic deploys

AI Analysis Core Code

```
// src/app/api/analyze-resume/route.ts export async function POST(req: NextRequest) { const
formData = await req.formData(); const file = formData.get('resume') as File; const
jobDescription = formData.get('jobDesc') as string; // Upload to Gemini File API const
fileManager = new GoogleAIFileManager(process.env.GEMINI_API_KEY!); const buffer =
Buffer.from(await file.arrayBuffer()); const upload = await fileManager.uploadFile(buffer, {
mimeType: 'application/pdf' }); // AI Analysis const model = genAI.getGenerativeModel({ model:
'gemini-2.0-flash', generationConfig: { responseMimeType: 'application/json' } }); const result
= await model.generateContent([ { fileData: { fileUri: upload.file.uri, mimeType:
'application/pdf' } }, `Analyze this resume for: ${jobDescription}. Return JSON with: score
(0-100), strengths[], weaknesses[], suggestions[], keyword_matches[], ats_score` ]); const
analysis = JSON.parse(result.response.text()); // Save to Firestore await addDoc(collection(db,
'analyses'), { userId: req.headers.get('x-user-id'), analysis, createdAt: new Date() }); return
NextResponse.json(analysis); }
```

Phase 10 — Pro Tips & Common Mistakes

Gemini Prompting Best Practices

- ✓ **System Instructions use karo:** Model ko ek role do: 'You are an expert code reviewer. Only respond in Hindi.'
- ✓ **Temperature control karo:** Creative tasks ke liye 0.9, factual ke liye 0.1 rakho
- ✓ **Few-shot examples do:** 2-3 input/output examples do prompt mein — accuracy badh jaati hai
- ✓ **JSON mode use karo:** Structured data ke liye response_mime_type='application/json' lagao
- ✓ **Context window use karo:** Gemini Flash ka 1M token context — poori codebase de sakte ho

Common Mistakes aur Solutions

✗ API key expose karo

Solution: NEVER .env.local mein rakho, client-side na use karo — server-side API routes banao

✗ Har request pe naya model init karo

Solution: Module level pe ek baar init karo, reuse karo

✗ Error handling nahi karna

Solution: Har AI call try-catch mein wrap karo — AI kabhi kabhi fail hota hai

✗ Rate limits ignore karna

Solution: Free tier mein 15 RPM limit hai — debounce aur caching lagao

✗ Ek hi badi function likhna

Solution: Small, focused functions banao — Jules aur Gemini easily samjhein

Rate Limit Handle Karna

```
// Simple retry with exponential backoff async function geminiWithRetry(prompt: string, retries = 3) { for (let i = 0; i < retries; i++) { try { return await askGemini(prompt); } catch (e: any) { if (e.status === 429 && i < retries - 1) { await new Promise(r => setTimeout(r, 2000 * (i + 1))); } else throw e; } } }
```

Costing — Free Limits Ka Summary

Tool	Free Limit	Kab Khatam Hoti Hai
Gemini Flash API	15 req/min, 1M tokens/day	Heavy production use pe
Firebase Firestore	50K reads, 20K writes/day	Large scale users pe
Firebase Storage	5 GB total	Agar bade files store karo
Vercel Hosting	100 GB bandwidth/month	Viral hone pe
Cloud Run	2M requests/month	Bahut zyada backend calls pe
Google Colab	Free GPU — ~12 hr sessions	Long training runs pe

Quick Reference — Cheatsheet

Useful Links

- Google AI Studio (Free Gemini): aistudio.google.com
- Gemini API Docs: ai.google.dev/docs
- Jules AI Coding Agent: jules.google.com
- Stitch UI Generator: labs.google/stitch
- NotebookLM: notebooklm.google.com
- Project IDX (Free IDE): idx.dev
- Firebase Console: console.firebase.google.com
- Google Cloud Console: console.cloud.google.com
- Gemini CLI GitHub: github.com/google-gemini/gemini-cli
- Gemma on Kaggle (Free GPU): kaggle.com/models/google/gemma

Complete 30-Day Roadmap

Timeline	Kya Karo
Week 1	Idea finalize (NotebookLM), tools setup, Next.js project create
Week 2	UI design (Stitch), auth (Firebase), basic pages ready
Week 3	Backend APIs, Gemini integration, database setup
Week 4	Advanced AI features, testing, Vercel deploy, demo ready

Yaad Rakho: Pehla version perfect nahi hoga — aur hona bhi nahi chahiye. Kaam karne wala product banaao, deploy karo, users se feedback lo, phir improve karo. Google ke ye tools tumhare liye sab kuch fast bana denge. Shuru karo!